

YI LU

(+86) 166-0522-6761 | tomlu@cs.toronto.edu | LinkedIn | GitHub | Google Scholar | Home Page

SUMMARY

- AI researcher focused on **LLM agents, agentic systems, tool-use post-training, retrieval-grounded reasoning, and multimodal ML**, with experience building agents that plan, call tools, retrieve evidence, verify outputs, and operate on real-world data.
- First/co-first author on ACL, CVPR Highlight, and ICLR Workshop Best Paper papers; experienced in driving research from problem formulation and system design to model training, evaluation, ablation, and paper writing.
- Hands-on experience with GRPO-style agentic RL, LoRA/PEFT/SFT, RAG/context management, dense retrieval, NL2SQL, tool-calling agents, SQL/Hive data pipelines, PyTorch, HuggingFace, vLLM, Docker/Linux, and GCP-based GPU development.

EDUCATION

University of Toronto

Sept. 2024 – Jan. 2026

Master of Science in Applied Computing, AI Concentration

GPA: 4.0/4.0 (All A/A+)

University of Manchester

Sept. 2020 – Jun. 2023

B.Sc. Hons Computer Science, Specialized in Artificial Intelligence

First Class Honours; WES GPA: 4.0/4.0; approximately top 5% among the cohort.

SELECTED RESEARCH AND APPLIED ML EXPERIENCE

Research Intern

Nov. 2025 – Apr. 2026

SKY Lab, Texas A&M University

Topic: LLM Agents, Long-Document Grounding, Review Evaluation

- Co-first authored **ReviewGrounder**, an **ACL 2026 Main paper**; the proposed agent system improves long-form academic reviews by grounding feedback in paper content, retrieved evidence, and paper-specific rubrics.
- Built **ReviewBench**, a content-centric evaluation framework that derives paper-specific rubrics from full papers, aggregated human reviews, and general meta-rubrics to assess coverage, evidence support, factual accuracy, and contradictory claims.
- Designed **ReviewGrounder** as a tool-integrated multi-agent refinement pipeline that separates drafting and grounding, using literature search, section-level analysis, evidence consolidation, and rubric-guided synthesis to produce more specific and evidence-supported long-form feedback.

Research Intern

Mar. 2025 – Nov. 2025

TIGER Lab, University of Waterloo

Topic: Agentic RL, Tool-Use Agents, LLM Post-training

- Co-first authored **VerlTool**, an open-source agentic RL framework built on **verl** for stateful, multi-turn, tool-calling LLM/MLLM agents; the technical report is accepted by TMLR 2026 and multiple ICLR 2026 workshops, including **Best Paper at SPOT**.
- Contributed to core framework design, extending RL with verifiable rewards from single-turn answer supervision to **Agentic RL with Tool Use**; supported custom tool services, agent-tool interaction logic, trajectory logging, reward/verifier design, multi-GPU training, and multi-task evaluation.
- Integrated Python coding, NL2SQL, and document-retrieval agent environments into VerlTool by adapting tool logic, reward design, and training recipes from systems such as SkyRL-SQL and Search-R1 into unified, reproducible training/evaluation recipes; debugged GRPO/DAPO-style rollout, reward computation, and tool-interaction failures.

Applied Research Intern (Co-op)

May 2025 – Dec. 2025

ModiFace, L'Oréal AI Lab

Topic: Agentic Animation Systems, Retrieval, Model Fine-tuning

- Led an applied Machine Learning project on an **agentic portrait-animation system** that turns user reference images and action descriptions into controllable character motion videos, unifying pose alignment, action template retrieval/LLM-based reranking, animation generation, quality verification, and retry policies into a **multi-agent orchestration system**.
- Designed the retrieval-and-reranking module for action selection, using dense retrieval over annotated motion descriptions followed by LLM reranking to select motion sequences that match user intent.
- Integrated verifier-driven quality control for pose alignment and generation artifacts, enabling dynamic retry and seed-adjustment policies. Fine-tuned an in-house portrait animation model and managed GCP multi-GPU experiments.

Research Intern

Sept. 2024 – May 2025

Opus AI Research

Topic: MLLM Reasoning, Video Understanding, Benchmarking

- First-authored **RSVP**, an **ACL 2025 Main paper** proposing a plug-and-play reasoning segmentation framework that improves open-vocabulary grounding via region-aware visual prompting and hierarchical multimodal reasoning.
- Contributed to **VEU-Bench**, a CVPR 2025 Highlight benchmark for high-level video editing understanding; worked on dataset construction, task repurposing, closed/open-source Vid-LLM evaluation, expert-model fine-tuning, analysis, and paper writing.
- Participated in CVPR 2025 video reasoning challenges on multilingual video understanding and complex video reasoning, achieving winner/runner-up results across two tracks.

Machine Learning Intern

Apr. 2024 – Sept. 2024

OpusClip

Topic: Video Editing Agents, Long-Video Retrieval

- Developed **Clip-Copilot**, an LLM-driven interactive video-editing agent for user-intent understanding, tool invocation, edit planning, state tracking, and iterative refinement; the prototype was later productized into the core product **Clip Anything**.
- Built **MM-Screenplay** / **MM-Screenplayer**, an agentic long-video understanding and retrieval framework that uses WhisperX, GPT-4, and Gemini to organize multi-modal evidence from hour-long videos for complex video QA.
- Applied the MM-Screenplayer pipeline to the CVPR 2024 LOVEU Workshop Long-Term VideoQA Challenge, contributing evidence organization and inference design for the winning solution.

Machine Learning Intern

Dec. 2023 – Mar. 2024

FaceMind

Topic: LLM Fine-tuning, RAG, Context Management

- Implemented a **long-term-memory RAG and context-management system** with Milvus and LangChain for locally deployable LLM chatbots, dynamically assembling user profiles, relevant prior dialogue, and retrieved memories into prompts for more consistent multi-turn conversations.
- Expanded retrievable chat history length by up to **200x**; fine-tuned LLaMA2-7B, Mixtral-8x7B, and Qwen-14B with LLaMA-Factory/QLoRA for persona control, response-style adaptation, and domain knowledge injection.

Data Science Intern

Aug. 2023 – Dec. 2023

ByteDance

Topic: Multimodal Understanding, Large-scale Data Science

- Fine-tuned a proprietary visual-language model with LoRA/PEFT for short-video commercial-intent understanding and deployed it into business-intent analysis workflows.
- Built large-scale short-video data and feature pipelines with Hive SQL/HDFS and LightGBM for commercial-value classification. Developed a zero-shot product-image matching pipeline with CLIP, OWL-ViT, YOLOv8, OpenCV, and Selenium-based web crawling.

SELECTED PUBLICATIONS

Improving Review Substantiveness with Rubric-Guided, Tool-Integrated Agents	ACL 2026 Main
VerlTool: Towards Holistic Agentic Reinforcement Learning with Tool Use	TMLR, ICLR 2026 SPOT Best Paper
RSVP: Reasoning Segmentation via Visual Prompting and Multi-Modal Chain-of-Thought	ACL 2025 Main
VEU-Bench: Towards Comprehensive Understanding of Video Editing	CVPR 2025 (Highlight)
Video Repurposing from User Generated Content: A Large-scale Dataset and Benchmark	AAAI 2025

SELECTED AWARDS

Multilingual Video Reasoning Evaluation Challenge @ CVPR 2025	Winner
Complex Video Reasoning and Robustness Evaluation Challenge @ CVPR 2025	Runner-up
Long-Term VideoQA Challenge, LOVEU Workshop @ CVPR 2024	Winner
Hour-long VideoQA Challenge, Second Perception Test Challenge @ ECCV 2024	Runner-up

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java
ML / Deep Learning: PyTorch, HuggingFace Transformers, PEFT/LoRA, SFT, adapter tuning, Scikit-learn, NumPy, Pandas
LLM / Agent Post-training: GRPO/DAPO-style agentic RL, RAG, tool-calling agents, NL2SQL, Agentic Systems
Systems / Cloud: Linux, Docker, Git, GCP, Azure, MySQL, Milvus, vLLM, LLaMA-Factory, ModelScope Swift
Data Systems: SQL, Hive SQL/HDFS, HuggingFace Datasets